Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To understand the process of convolution over the image and apply over the classification problem**	
Experiment No: 4	**Date:19/03/2026**	**Enrolment No:92301733054**

Aim: To understand the process of convolution over the image and apply over the classification problem

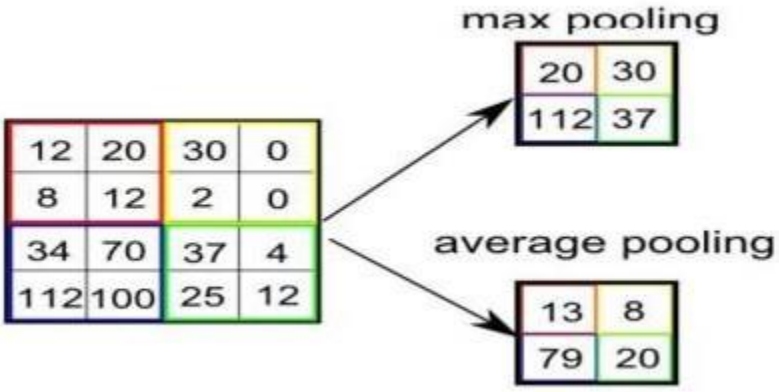
IDE: Google Colab

Theory:

Convolutional Neural Networks (CNN) are complex feed forward neural networks. CNNs are used for image classification and recognition because of its high accuracy There are three types of layers in a convolutional neural network: i. Convolutional layer ii. Pooling layer iii. Fully connected layer Each of these layers has different parameters that can be optimized and performs a different task on the input data.

What is Pooling Layer?

Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data through dimensionality reduction. There are two types of Pooling i. Average Pooling. ii. Max Pooling Max Pooling returns the maximum value from the portion of the image covered by the Kernel. On the other hand, Average Pooling returns the average of all the values from the portion of the image covered by the Kernel. Max Pooling also performs as a Noise Suppressant. It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction. On the other hand, Average Pooling simply performs dimensionality reduction as a noise suppressing mechanism. Hence, we can say that Max Pooling performs a lot better than Average Pooling.



What is Convolutional Layer?

Convolutional layers are the major building blocks used in convolutional neural networks. A convolution is the simple application of a filter to an input that results in an activation. Repeated application of the same filter to an input results in a map of activations called a feature map, indicating the locations and strength of a detected feature in an input, such as an image. A convolutional layer contains a set of filters whose parameters need to be learned. The height and weight of the filters are smaller than those of the input volume. Each filter is convolved with the input volume to compute an activation map made of neurons.

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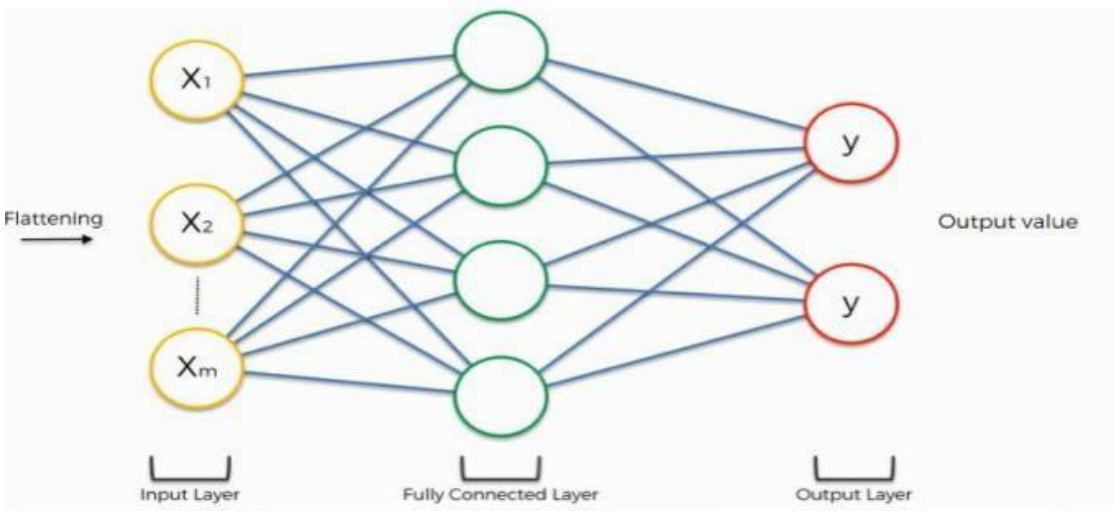
What is Fully Connected Layer?

A fully connected layer that takes the output of convolution/pooling and predicts the best label to describe the image We have three layers in the full connection step i. Input layer ii. Fully-connected layer iii. Output layer

Input Layer: It takes the output of the previous layers, "flattens" them and turns them into a single vector that can be an input for the next stage.

Fully Connected Layer: It takes the inputs from the feature analysis and applies weights to predict the correct label.

Output Layer: It gives the final probabilities for each label.



ReLU Layer: ReLU is an activation function. Rectified Linear Unit (ReLU) transform function only activates a node if the input is above a certain quantity, while the input is below zero, the output is zero, but when the input rises above a certain threshold, it has a linear relationship with the dependent variable. The main aim is to remove all the negative values from the convolution. All the positive values remain the same but all the negative values get changed to zero.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Normalize the data
5. Pre-process the data
6. Visualize the Data
7. Write the CNN model function
8. Write the Cost Function

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9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Observe the cost function vs iterations learning curve

Program (Code):

```
#import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import keras
from keras.datasets import fashion_mnist
import keras.models as models
import keras.layers as layers
from keras import regularizers
from keras.layers import Dropout
import tensorflow as tf
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense,Activation,Flatten,Dropout,Conv2D,
MaxPooling2D, LeakyReLU
from tensorflow.keras.utils import to_categorical   #for the categorical data
print(keras.__version__)
#import the dataset #x: Images and Y: labels of the images
(train_X,train_Y),(test_X,test_Y)=fashion_mnist.load_data()
plt.imshow(train_X[0])
print(train_Y[0])
print(train_X.shape,train_Y.shape)
print(test_X.shape, test_Y.shape)
classes=np.unique(train_Y)
nClasses=len(classes)
print(classes)
print(nClasses)
train_X=train_X.reshape(-1,28,28,1)
test_X=test_X.reshape(-1,28,28,1)
print(train_X.shape,train_Y.shape)
print(test_X.shape, test_Y.shape)
#converting the values of the pixels from 0-1
#type-conversion
train_X=train_X.astype('float32')
test_X=test_X.astype('float32')
train_X=train_X/255
test_X=test_X/255
train_Y_one_hot=to_categorical(train_Y)
test_Y_one_hot=to_categorical(test_Y)
train_Y[0]
```

```
train_Y_one_hot[0]
from sklearn.model_selection import train_test_split
train_X,valid_X,train_label,valid_label=train_test_split(train_X,train_Y_one_hot,test_s
ize=0.2,random_state=13)
train_X.shape,valid_X.shape,train_label.shape,valid_label.shape
batch_size=64
epochs=20
num_classes=10
model=tf.keras.Sequential()
model=tf.keras.Sequential()
#1st hidden layer
model.add(tf.keras.layers.Conv2D(32,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.laymodel.add(tf.keras.layers.Flatten())
ers.MaxPooling2D(pool_size=(2,2),padding='same'))
#2nd hidden layer
model.add(tf.keras.layers.Conv2D(64,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
#3rd hidden layer
model.add(tf.keras.layers.Conv2D(128,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
model.add(tf.keras.layers.Dense(128,activation='linear'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.Dense(num_classes,activation='softmax'))#output final layer
model.compile(loss=tf.keras.losses.categorical_crossentropy,optimizer=tf.keras.optimize
rs.Adam(),metrics=['accuracy'])
model_train=model.fit(train_X,train_label,batch_size=64,epochs=20,verbose=1,validation_
data=(valid_X,valid_label))
testing_evaluation=model.evaluate(test_X,test_Y_one_hot)
model_train.history
accuracy=model_train.history['accuracy']
val_accuracy=model_train.history['val_accuracy']
loss=model_train.history['loss']
val_loss=model_train.history['val_loss']
epochs=range(len(accuracy))
plt.plot(epochs,accuracy,label='training accuracy')
plt.plot(epochs,val_accuracy,label='validation accuracy')
plt.title('evaluating based on accuracy')
plt.legend()
plt.show()
plt.plot(epochs,loss,label='training loss')
plt.plot(epochs,val_loss,label='validation loss')
```

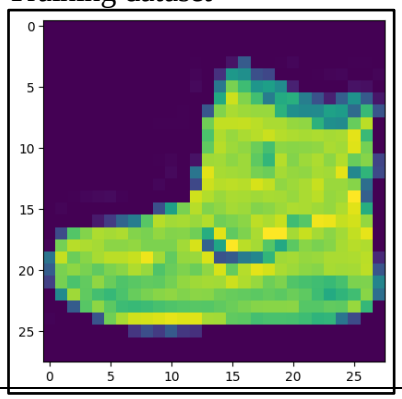
```
plt.title('evaluating based on loss')
plt.legend()
plt.show()
regmodel=tf.keras.Sequential()
#1st hidden layer
regmodel.add(tf.keras.layers.Conv2D(32,(3,3),activation="linear",padding='same'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.25))
#2nd hidden layer
regmodel.add(tf.keras.layers.Conv2D(64,(3,3),activation="linear",padding='same'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.25))
#3rd hidden layer
regmodel.add(tf.keras.layers.Conv2D(128,(3,3),activation="linear",padding='same'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.40))

regmodel.add(tf.keras.layers.Flatten())
regmodel.add(tf.keras.layers.Dense(128,activation='linear'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.Dropout(0.3))
regmodel.add(tf.keras.layers.Dense(num_classes,activation='softmax'))#output final
layer
regmodel.compile(loss=tf.keras.losses.categorical_crossentropy,optimizer=tf.keras.optim
izers.Adam(),metrics=['accuracy'])
reg_model_train=regmodel.fit(train_X,train_label,batch_size=64,epochs=20,verbose=1,vali
dation_data=(valid_X,valid_label))
```

Results:

To be attached with

a. Training dataset



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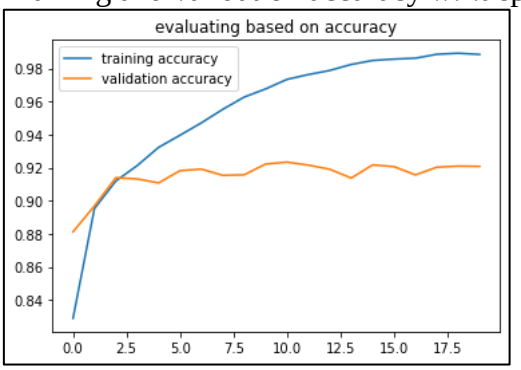
b. Model summary

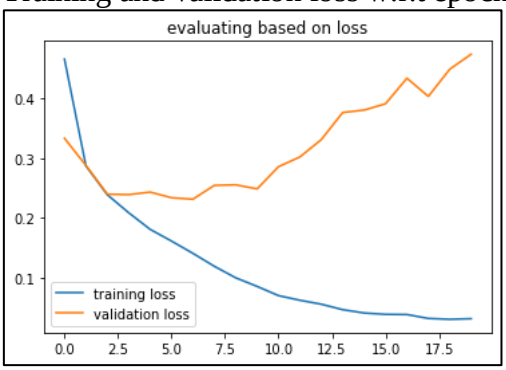
```
print(train_X.shape,train_Y.shape)
print(test_X.shape, test_Y.shape)
```

```
(60000, 28, 28, 1) (60000,)
(10000, 28, 28, 1) (10000,)
```

c. Training and validation accuracy w.r.t epochs before regularization



d. Training and validation loss w.r.t epochs before regularization



Pre Lab Exercise :

a. What is Convolution process? Explain with example.

- Convolution is the process of applying a filter (kernel) over an image to extract important features such as edges, textures, or patterns.

Example:

- A 3×3 filter moves over an image and performs element-wise multiplication and summation, producing a feature map.

b. What are different layers in CNN model?

The main layers in CNN are:

1. Convolutional Layer – Extracts features from input image

2. Activation Layer (ReLU) – Introduces non-linearity
3. Pooling Layer – Reduces dimensions
4. Fully Connected Layer – Performs classification

c. What is the requirement of pooling layer in CNN?

Pooling layer is used to:

- Reduce spatial dimensions
- Decrease computational cost
- Prevent overfitting
- Extract dominant features

d. Why ReLU activation is used after convolution?

- Removes negative values
- Introduces non-linearity
- Helps faster training
- Avoids vanishing gradient problem

<u>Observation and Result Analysis:</u>

e. Nature of Dataset

- Dataset used is Fashion MNIST
- Contains grayscale images (28×28 pixels)
- Total 10 classes (clothing items)
- Data is normalized between 0 and 1
- Suitable for image classification tasks

f. Training Process without Regularization

- Model learns patterns using convolution + pooling layers
- Validation accuracy may fluctuate
- Overfitting may occur:
  - Training accuracy high

- Validation accuracy lower

g. After Regularization in Training Process

- Dropout layers are added
- Overfitting is reduced
- Training accuracy slightly decreases
- Validation accuracy improves
- Model generalizes better

h. Observation over Learning Curves

- Accuracy graph:
  - Training accuracy increases steadily
  - Validation accuracy stabilizes
- Loss graph:
  - Training loss decreases
  - Validation loss reduces but may fluctuate

<u>Post Lab Exercise :</u>

a. Why CNN is preferred over ANN for images?

- CNN preserves spatial information of images
- Uses fewer parameters than ANN
- Automatically extracts features (edges, shapes)
- More efficient and accurate for image tasks

b. Can CNN be applied over text data?

Yes ,

- Text is converted into numerical form (embeddings)
- CNN applies filters over sequences
- Used in:
  - Sentiment analysis

- Text classification

c. What is the role of Dropout layer?

- Randomly disables neurons during training
- Prevents overfitting
- Improves model generalization

d. What happens if max pooling is replaced with min pooling?

- Min pooling selects smallest value
- Important features may be lost
- Model performance decreases
- Noise may increase